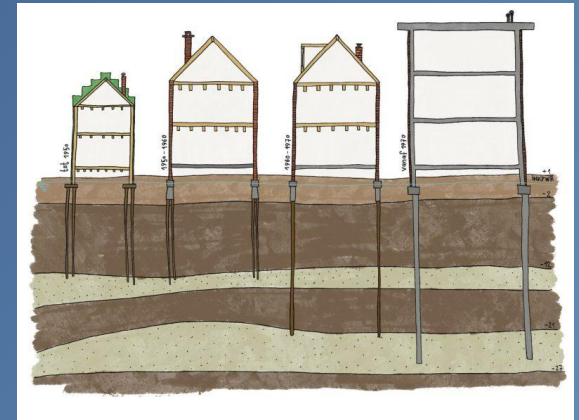


From Code to Confidence: Toolchain Qualification in ITS

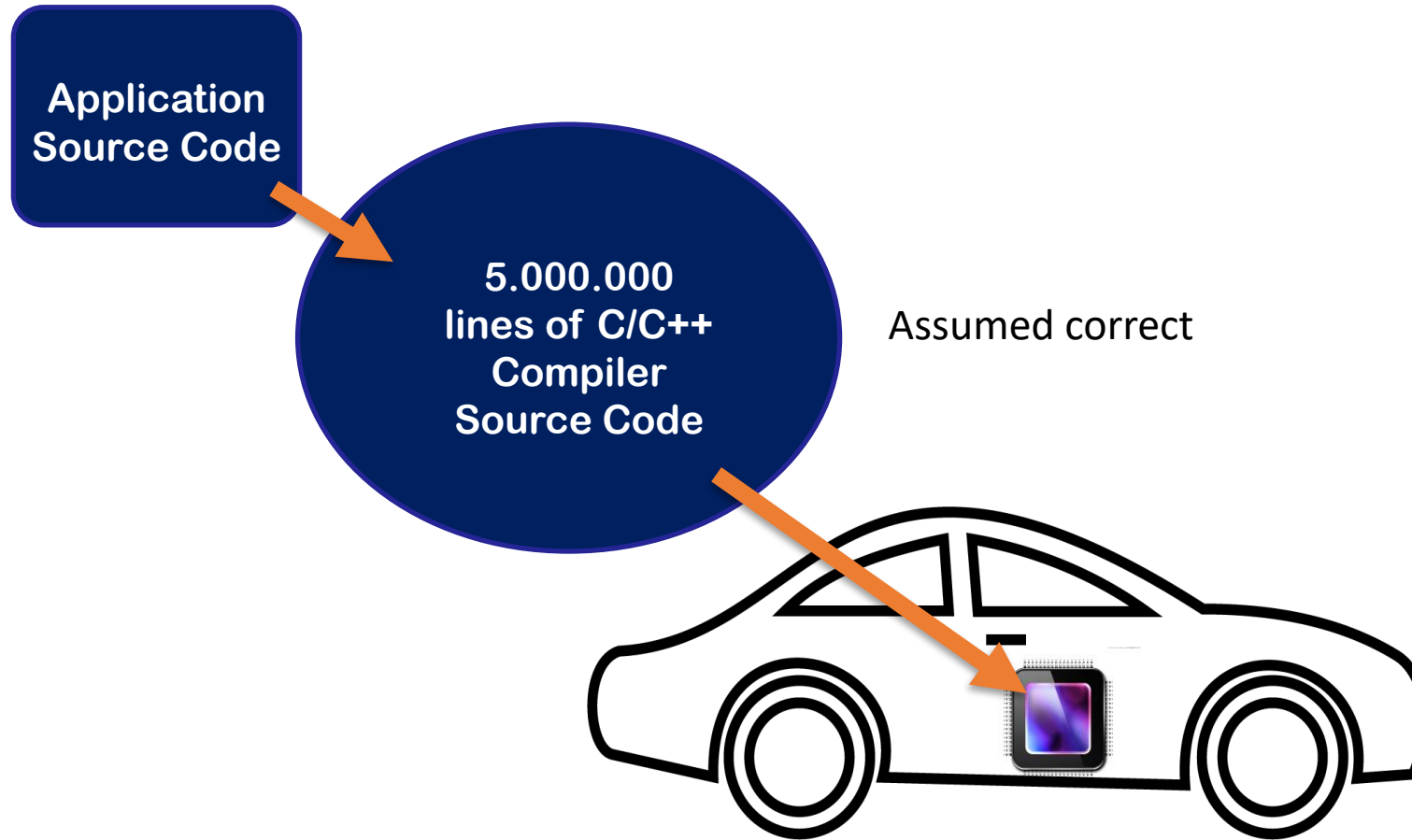
Sjoerd van der Zwaan (Solid Sands)



A System of Systems

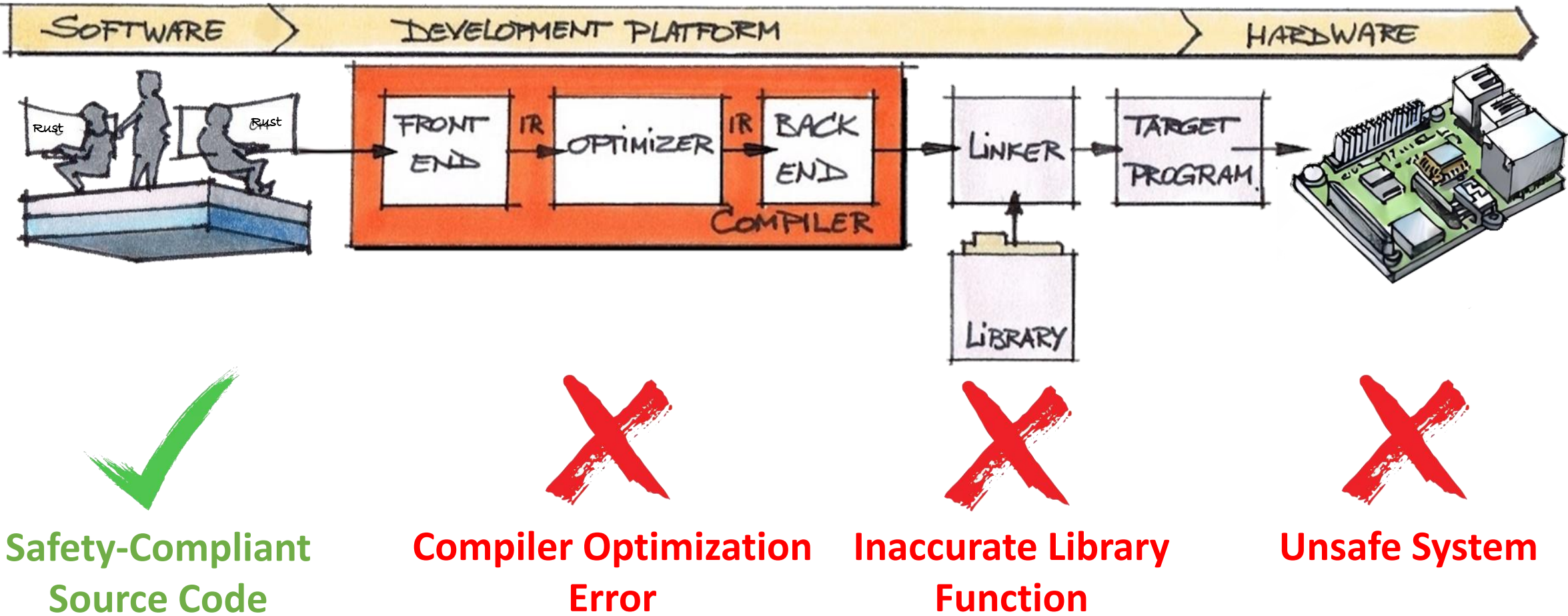


The Toolchain as Invisible Foundation



The toolchain sits between engineering intent and executable reality

The Silent Risk



Examples of Compiler Errors

- Compiler Bug for absolute values

```
int8 x = -3;  
Out = abs(x);
```



Expected: **3**
Got: **-3**

- Dead code optimization for non-dead code

```
Int myFun(int in1)  
{  
    return Out * 2;  
}  
  
/*In other Function*/  
Out = 4;  
Out = myFun(2);
```



Expected: **Out = 8**,
Got: **Out = 4**

“Out=4;” is not dead code but was removed by the compiler as such

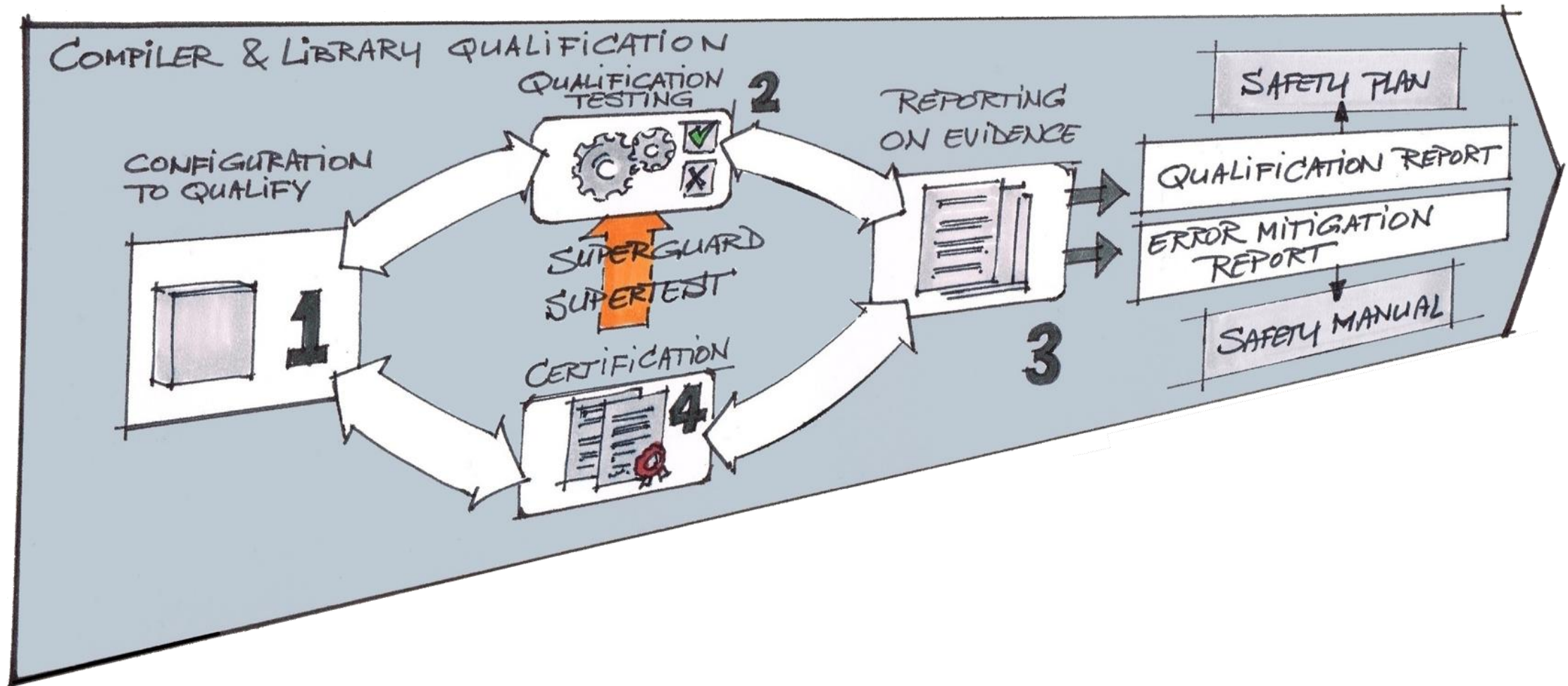
Why ITS Amplifies the Toolchain Risk

- many processors/architectures
- multiple suppliers
- C/C++/Rust and other languages
- common compilers and libraries
- common build infrastructure
- repeated software components
- safety functions distributed across subsystem



One systematic toolchain issue can propagate across many parts of a system-of-systems.

Building Trust with Evidence



The Update Paradox

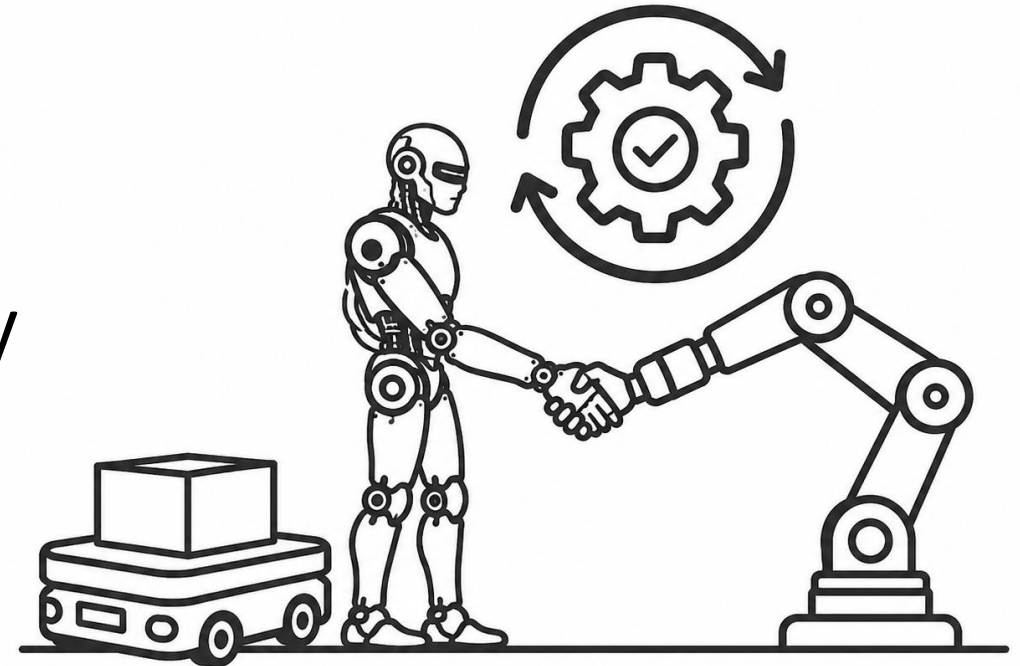
- Freeze
“stable qualification baseline, predictable behaviour, BUT vulnerable to security issues and difficult to adopt new platforms/features”
- Flow
“security updates, bug fixes, new hardware, new compiler versions, new language features, BUT qualification evidence must evolve”



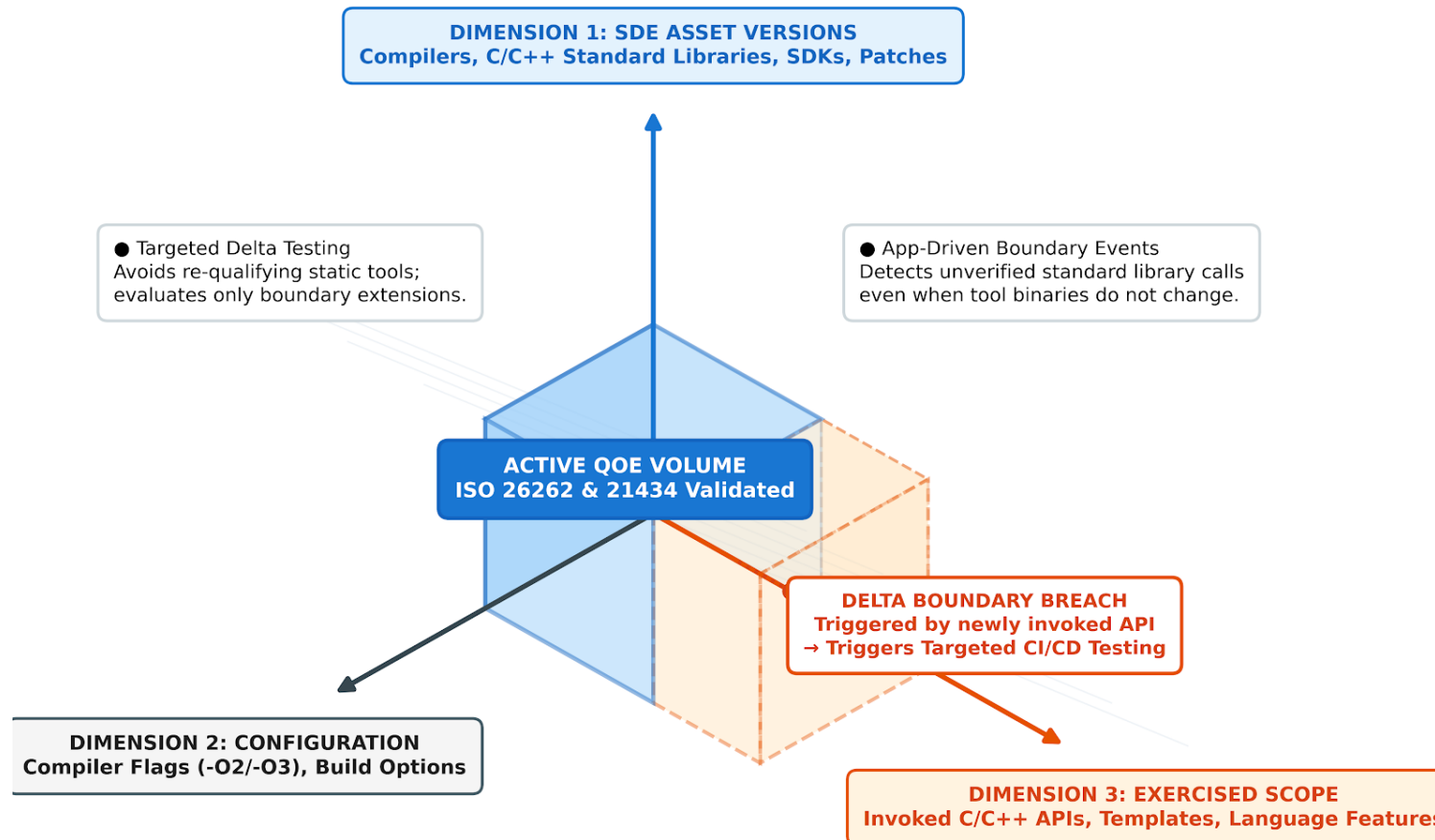
How do we make “Flow” safe?

The Shift to Continuous Qualification

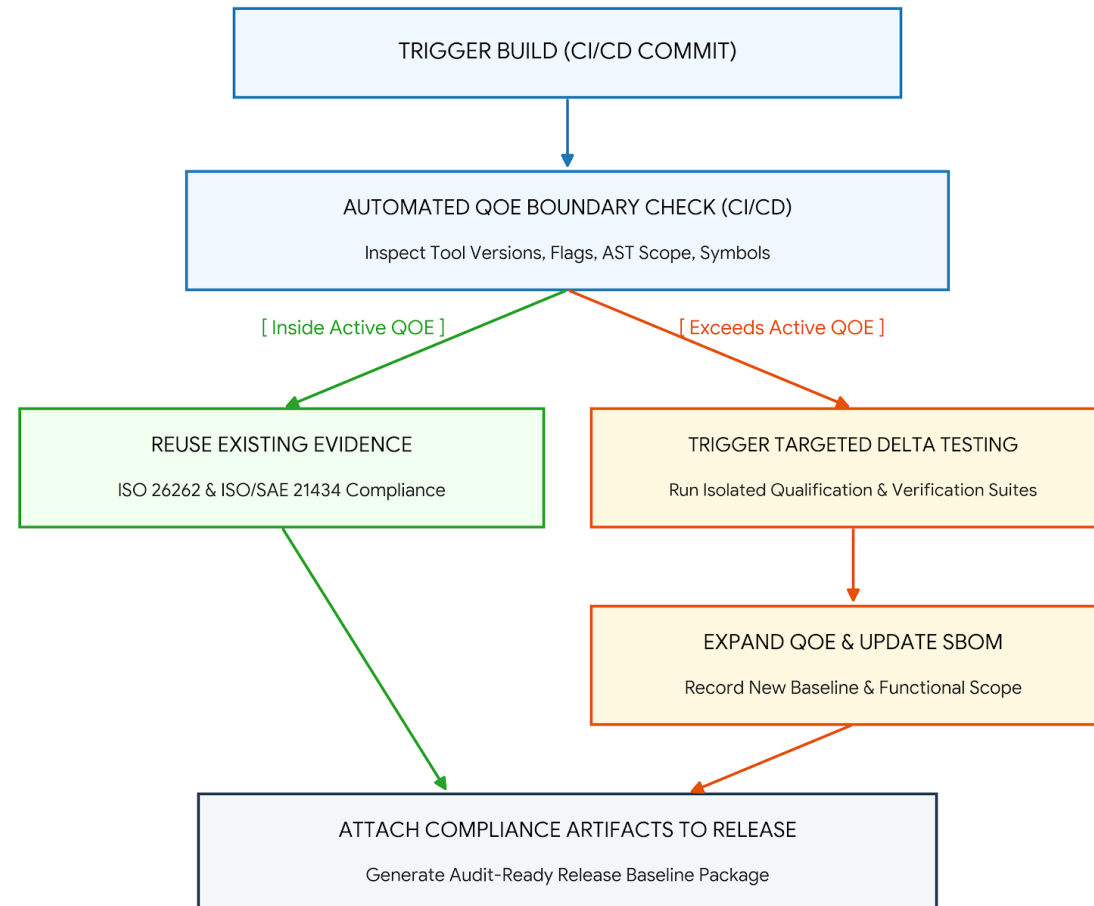
- Qualification is no longer a one-time event
- It becomes a continuous engineering activity
- Toolchain changes are continuously evaluated
- Confidence is actively maintained over time



The Qualified Operating Envelope



Continuous Qualification Framework



Regulation is Catching Up

- Software becomes a regulated system component
- Toolchains fall inside compliance scope
- Evidence replaces assumptions
- Continuous verification is expected



EU AI Act
EU Cyber Resilience Act
EU Machinery Regulation

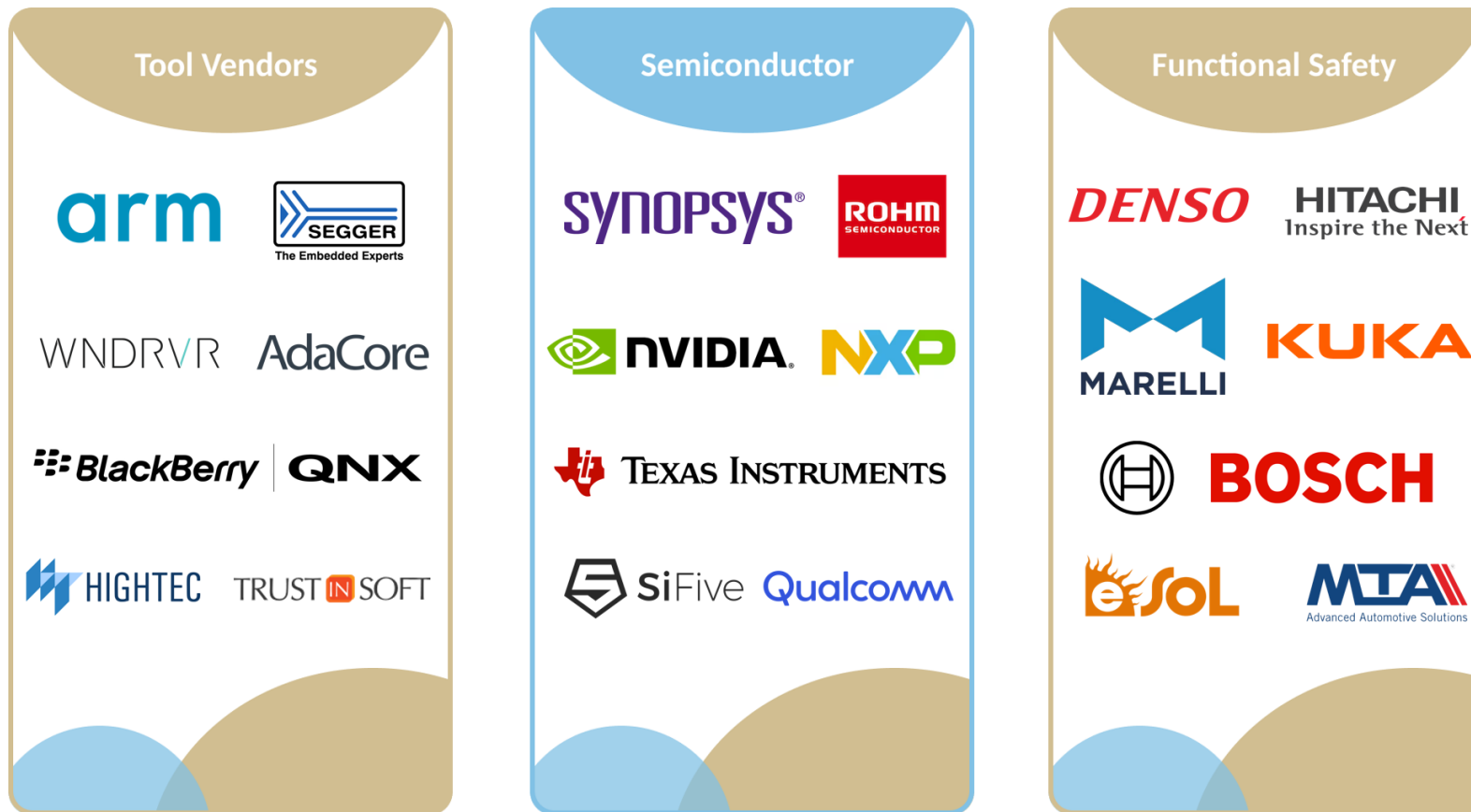
Why it Matters Strategically

- faster technology adoption
- faster security response
- reduced requalification effort
- greater reuse of evidence
- better supplier transparency
- shorter change-impact analysis
- more defensible safety cases
- greater confidence in software updates

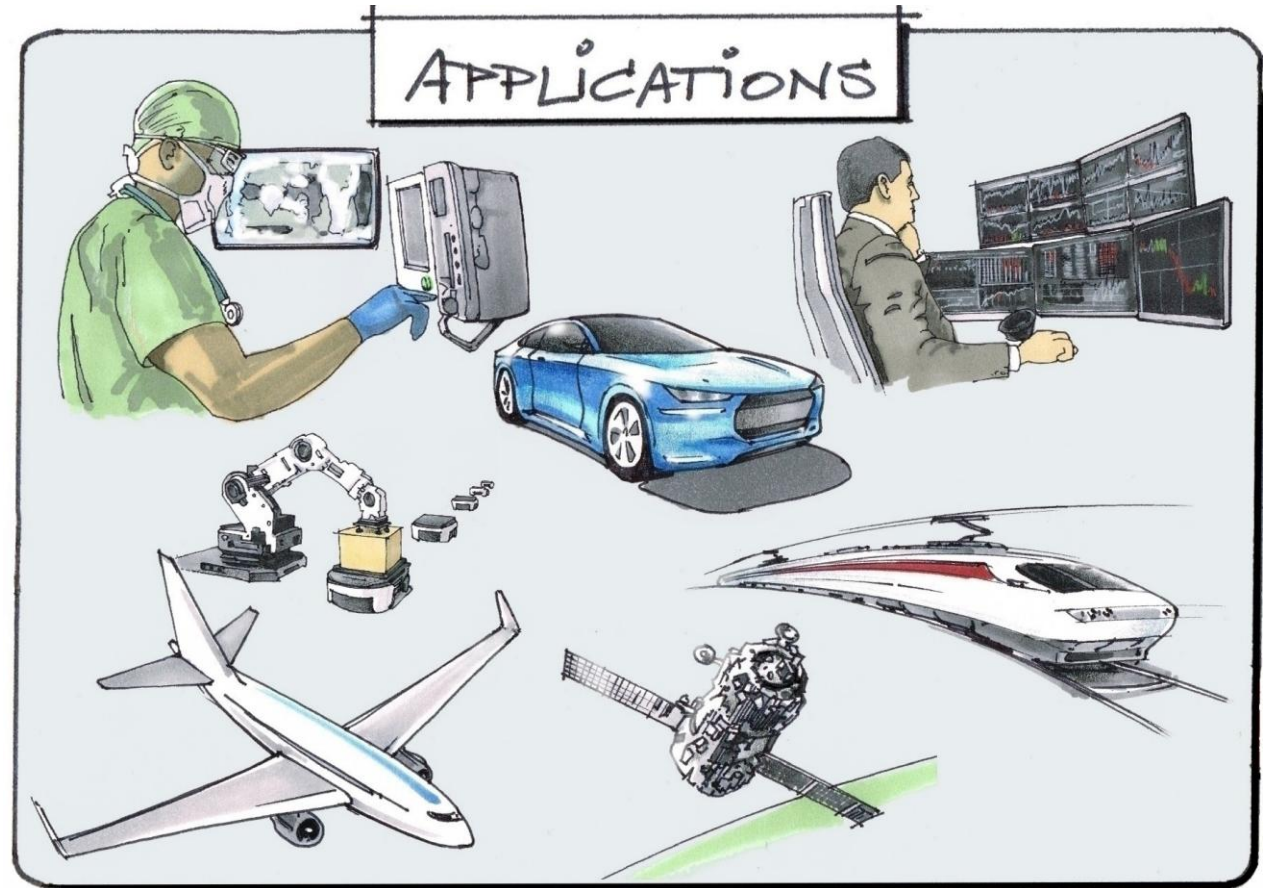
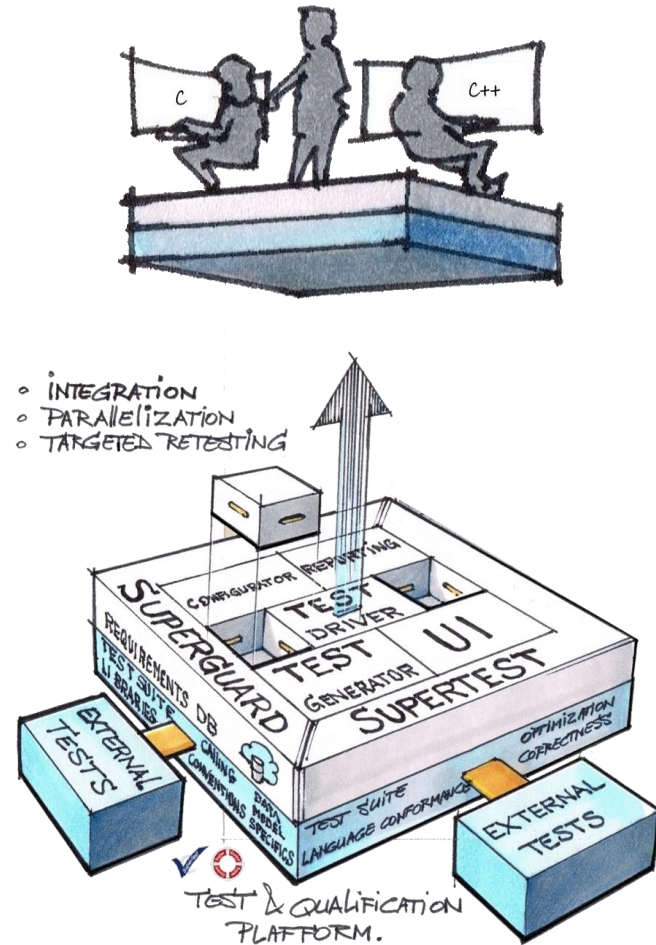
The ability to change safely becomes a strategic capability.

Building an Ecosystem of Trust

Toolchain Confidence Cannot Be Built by One Company Alone



The Role of Solid Sands



From Toolchain to Unified Toolchain

- Modern safety-critical systems increasingly combine multiple languages. Qualification and assurance should not become fragmented across separate toolchains.
- Focus on consistency of evidence and engineering workflow.



Continuous Qualification Made Practical

- Make the envelope explicit
- Detect relevant drift
- Refresh targeted evidence



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